

Note that the character string type is a bounded sequence of characters whose size bound format is not specified in the data model, but rather a property of the underlying encoding. Therefore, no assumptions are to be made about the presence or absence of a length prefix or NULL-terminator byte, or other implementation considerations.

It is RECOMMENDED to define constraints on the maximum possible string length.

If at least one of the code points within the string has value 31 (0x1F), which is Unicode **INFORMATION SEPARATOR 1** and ASCII **Unit Separator**, then any client making use of the string SHALL only consider the code points that appear before the first **INFORMATION SEPARATOR 1** as being the textual information carried by the string. Any comparison between such a string and other strings SHALL use the textual component before the first **INFORMATION SEPARATOR 1**. The remainder of the character string after a first **INFORMATION SEPARATOR 1** is reserved for future use by this specification. Implementations of this version of the specification SHALL NOT produce character strings containing **INFORMATION SEPARATOR 1**.

7.19.2.41. IP Address Types

Either an IPv4 or an IPv6 address as defined below.

7.19.2.42. IPv4 Address Type

The IPv4 address data type is derived from an octet string. The octets SHALL correspond to the four octets in network byte order that comprise an IPv4 address represented utilizing quad-dotted notation.

Examples of encoding:

- Address 192.168.2.235 → C0A802EB
- Address 10.4.200.75 → 0A04C84B

7.19.2.43. IPv6 Address

The IPv6 address data type is derived from an octet string. The octets SHALL correspond to the full 16 octets that comprise an IPv6 address as defined by RFC 4291. The octets SHALL be presented in network byte order.

Examples of encoding:

- Address 2001:DB8:0:0:8:800:200C:417A → 20010DB8000000000000080800200C417A
- Address 2001:0DB8:1122:3344:5566:7788:99AA:BBCC → 20010DB8112233445566778899AABBCC

7.19.2.44. IPv6 Prefix

The IPv6 prefix data type is derived from an octet string. The octets SHALL be encoded such that:

- The first octet SHALL encode the prefix length, in bits, in the range of 0 to 128.
 - A value of 0 indicates an absent/invalid prefix.
- The subsequent octets SHALL encode the contiguous leftmost bits of the prefix, in network byte